SELECT within SELECT Tutorial

Language:	English • 日本語 • 中文

This tutorial looks at how we can use SELECT statements within SELECT statements to perform more complex queries.

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000
...				

Using nested SELECT

Summary

Contents

Bigger than Russia

List each country name where the population is larger than that of 'Russia'.

```
world(name, continent, area, population, gdp)
```

```
select name from world where population>(select population from world where name='Russia')
```

Submit SQL restore default

result

0:00 / 4:32

Richer than UK

Show the countries in Europe with a per capita GDP greater than 'United Kingdom'.

Per Capita GDP

```
select name from world where continent='Europe' and gdp/population>(select gdp/population from
world where name = 'United Kingdom');
```

Submit SQL

restore default

result

0:00 / 3:36

Neighbours of Argentina and Australia

List the name and continent of countries in the continents containing either Argentina or Australia. Order by name of the country.

```
select name,continent from world
where continent=any(select continent from world
where name='Argentina' or name ='Australia'
) order by name
;
```

Submit SQL restore default

result

Between Canada and Poland

Which country has a population that is more than United Kingdom but less than Germany? Show the name and the population.

```
select name, population from world where population>(select population from world where
name='United Kingdom') and population<(select population from world where name='Germany')
```

Submit SQL

restore default

result

Percentages of Germany

Germany (population roughly 80 million) has the largest population of the countries in Europe. Austria (population 8.5 million) has 11% of the population of Germany.

Show the name and the population of each country in Europe. Show the population as a percentage of the population of Germany.

The format should be *Name, Percentage* for example:

name	percentage
Albania	3%
Andorra	0%
Austria	11%
...	...

Decimal places
Percent symbol %

```
select name,
concat(round((population/(select population from world where name='Germany')*100)) ,'%') as
percentage
from world
```

Submit SQL

restore default

result

0:00 / 9:14

To get a well rounded view of the important features of SQL you should move on to the next tutorial concerning aggregates.

To gain an absurdly detailed view of one insignificant feature of the language, read on.

We can use the word `ALL` to allow >= or > or < or <=to act over a list. For example, you can find the largest country in the world, by population with this query:

```
SELECT name
  FROM world
 WHERE population >= ALL(SELECT population
                           FROM world
                          WHERE population>0)
```

You need the condition **population>0** in the sub-query as some countries have **null** for population.

Bigger than every country in Europe

Which countries have a GDP greater than every country in Europe? [Give the name only.] (Some countries may have NULL gdp values)

```
select name from world
where gdp>all(select gdp from world where continent = 'Europe' and gdp is not null)
```

Submit SQL

restore default

result

We can refer to values in the outer SELECT within the inner SELECT. We can name the tables so that we can tell the difference between the inner and outer versions.

Largest in each continent

Find the largest country (by area) in each continent, show the continent, the name and the area:

```
where area=any(select max(area) from world group by continent)
correct but when the outter area value are same then it gives error
so do not use this
*/


select continent,name,area from world as w where area=(select max(area) from world where
continent=w.continent)
```

Submit SQL restore default

result

The above example is known as a **correlated** or **synchronized** sub-query.

Using correlated subqueries

First country of each continent (alphabetically)

List each continent and the name of the country that comes first alphabetically.

```
/*
select name from world
where continent='Europe' order by name limit 1
*/


select continent, min(name) from world group by continent;
```

Submit SQL restore default

result

Difficult Questions That Utilize Techniques Not Covered In Prior Sections

Find the continents where all countries have a population <= 25000000. Then find the names of the countries associated with these continents. Show name, continent and population.

```
/*select max(population) from world where continent='Asia'*/

select name, continent, population from world as w
where 25000000>= (select max(population) from world where continent=w.continent)
```

Submit SQL

restore default

result

Three time bigger

Some countries have populations more than three times that of all of their neighbours (in the same continent). Give the countries and continents.

```
--

select name,continent from world as w
where population>(select 3*max(population) from world where continent=w.continent and
name!=w.name)

/*
select 3*max(population) from world where continent=w.continent
```

Submit SQL

restore default

result

Nested SELECT Quiz